AI CYBER SABOTAGE

Task 4 - High-Resolution Developer Storyboard

Three-round Red vs Blue game. Every active teammate votes privately. Majority vote becomes one team action. Attack cards form a shared single-use arsenal; Defence cards are reusable. Learning Mode observes only.

ATTACK	DEFENCE	LEARNING MODE
6-card SHARED arsenal **Single-use** **Used card becomes SPENT for both teams**	**6-card response deck** **Reusable every defence round** **Subject to remaining team budget**	**Observer only** **No voting / scoring / team assignment** **Tablet stays locked**

Round flow: R1 Red attacks / Blue defends -> R2 Blue attacks / Red defends -> R3 Red attacks / Blue defends -> team result -> payout -> live leaderboard -> Ultimate Showdown (3 questions).

01. NON-NEGOTIABLE GAME RULES

Implementation rules agreed during review.

Participation	Only players with status COMPETING at Task 4 start are assigned Red / Blue.
Learning Mode	Observer-only for entire Task 4. No cards, votes, team points or interaction.
Team split	Split active competitors as evenly as possible.
Voting	Every active teammate has a live tablet and privately selects one card.
Team action	Highest-voted card becomes ONE team action; cost deducted ONCE.
Vote changes	Players may change selection until timer expires; last choice counts.
Live totals	Hidden until voting closes.
Tie	5-second re-vote using tied cards only; second tie -> lower-cost card.
Budgets	Red and Blue each start at $100k. Attack + defence spend from that team's budget.
Attack arsenal	6 attacks are SHARED across both teams and SINGLE-USE. Once played, that card is SPENT for the rest of Task 4.
Spent attack UI	Keep card visible but disabled: SPENT - USED BY RED TEAM / BLUE TEAM.
Attack availability	Round 1: 6 available -> Round 2: 5 -> Round 3: 4, assuming one is spent each round.
Defence deck	All 6 Defence cards reappear every defence round. Defence cards are NOT spent.
Affordability	Any attack or defence above the acting team's remaining budget stays visible but disabled: NOT ENOUGH BUDGET.

02. TEAM REVEAL - 10 SECONDS

20 active competitors would typically split 10 Red / 10 Blue.

HOST SCREEN

TASK 4: AI CYBER SABOTAGE
LOOK AT YOUR TABLET.
Find your team.
10 seconds.

PLAYER TABLET

COMPETING: **YOU ARE RED TEAM / YOU ARE BLUE TEAM**
Find your team.
Shared team budget: **$100,000**

LEARNING: **YOU ARE IN LEARNING MODE**
Observe on Host screen.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Assign only competing players; Learning Mode enters observer_lock.	Initialize team budgets=100000, scores=0, shared_attack_spent=[]

03. TEAM VOTING - MANY INPUTS, ONE ACTION

All teammates participate; there is no captain tablet.

HOST SCREEN

10 Red players are voting.
Totals hidden while timer runs.
At zero: **RED TEAM HAS SPOKEN**
Winning card + vote count + cost are revealed.

PLAYER TABLET

Six eligible cards shown.
Tap ONE. Change choice until timer expires.
Your current choice is visibly selected.
Waiting for your team...

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Store one current_selection per eligible player. Aggregate at timeout.	Majority = one team action; charge once. Tie -> 5 sec re-vote -> lower-cost if tied again.

04. CARD LOGIC - SHARED ATTACK ARSENAL / REUSABLE DEFENCE

Attack and Defence intentionally behave differently.

ATTACK: shared across Red + Blue and single-use. Once any team plays a method it becomes SPENT and cannot be selected again. Keep it visible, greyed/disabled, and label who used it.

ATTACK - SINGLE USE	DESCRIPTION	COST	SUCCESS	FAIL
Deepfake Video	Convincing fake video of someone the victim trusts	**$30k**	**+900**	**-350**
Cloned Voice	Copy someone's voice for a fake call / message	**$25k**	**+750**	**-300**
Personalised Message	Use personal/context details to make a message believable	**$20k**	**+650**	**-250**
AI-Written Message	Make a scam message polished and convincing	**$10k**	**+300**	**-100**
Fake Digital Identity	Create a convincing fake profile / persona	**$20k**	**+600**	**-250**
AI Agent Attack	Automate actions and adapt the attack	**$35k**	**+1,000**	**-400**

DEFENCE: reusable in every defence round. A response can be selected again later, provided the defending team can afford it.

DEFENCE - REUSABLE	DESCRIPTION	COST	SUCCESS	FAIL
Stop the Action	Pause the request/action before anything happens	**$20k**	**+750**	**-150**
Verify Independently	Confirm through a trusted source/channel	**$18k**	**+700**	**-150**
Report & Escalate	Flag suspicious activity through the right process	**$15k**	**+600**	**-175**
Investigate Further	Inspect content/output for signs something is wrong	**$8k**	**+300**	**-275**
Ask Someone Else	Get a colleague's opinion before deciding	**$5k**	**+150**	**-350**
Continue Carefully	Proceed but avoid sharing anything sensitive	**$0**	**+50**	**-500**

05. ROUND 1 - RED ATTACK

Personalised phishing. All six Attack methods are initially available.

HOST SCREEN

ROUND 1 / 3
RED TEAM - ATTACK
Make a phishing email convincing enough to drive an employee to a fraudulent login page.
Choose ONE AI capability.
20 sec | Red $100k

PLAYER TABLET

ALL RED TABLETS
6 Attack cards available unless unaffordable.
Private vote -> majority -> one attack.
Winning attack becomes **SPENT** for BOTH teams after play.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Run attack_vote(team=Red). Blue waits; Learning observes.	Deduct winning cost once; add attack_id to shared_attack_spent. Round 2 must disable it globally.

06. ROUND 1 - BLUE DEFENDS

Blue responds using the reusable Defence deck.

HOST SCREEN

ATTACK INCOMING
Email knows your name, role, manager and project.
Urgent login requested.
BLUE: CHOOSE YOUR RESPONSE.
20 sec | Blue $100k

PLAYER TABLET

ALL BLUE TABLETS
All 6 Defence cards reappear.
Cards over Blue's budget are disabled.
Private vote -> majority -> one defence.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Run defence_vote(team=Blue). Defence cards do not enter spent state.	Deduct defence cost once. Resolve pairing; apply rewards/penalties.

07. ROUND 1 - RESOLVE

The attack is now permanently removed from the shared arsenal.

HOST SCREEN

ROUND 1 RESULT
Show selected Red attack vs Blue defence.
BLOCKED / SUCCESSFUL
Update points + budgets.
Attack card status: **SPENT - USED BY RED TEAM**.

PLAYER TABLET

Both teams see outcome, remaining budget and running team score.
Learning Mode remains static observer.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Persist round result and spent attack ID.	Round 2 Attack UI loads 5 selectable attacks + 1 visible SPENT card.

08. ROUND 2 - BLUE ATTACK

CEO impersonation. Blue inherits the depleted shared arsenal.

HOST SCREEN

ROUND 2 / 3
BLUE TEAM - ATTACK
Convince Finance that the CEO urgently needs an unusual payment approved.
20 sec | Blue budget $XXk

PLAYER TABLET

BLUE TABLETS
5 attack methods remain selectable.
Round 1 method stays visible but disabled: **SPENT - USED BY RED**.
Any unaffordable card: **NOT ENOUGH BUDGET**.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Filter shared attack arsenal by spent state + Blue affordability.	Winning Blue attack is charged once and added to shared_attack_spent. Round 3 now has 4 unspent attacks.

09. ROUND 2 - RED DEFENDS

Defence does not deplete.

HOST SCREEN

ATTACK INCOMING
Voice/video/message appears to come from the CEO.
Urgent unusual payment request.
RED: CHOOSE YOUR RESPONSE.

PLAYER TABLET

RED TABLETS
Same 6 reusable Defence cards.
Verify Independently is strong for identity impersonation.
Only affordability can disable a Defence card.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Run same defence component; no spent flag on defence cards.	Resolve; update budgets/scores. Learning: familiar voice/face is not proof of identity.

10. ROUND 2 - RESOLVE

Two Attack cards are now spent globally.

HOST SCREEN

ROUND 2 RESULT
BLUE attack vs RED defence.
Outcome + point deltas.
Two attack methods now marked SPENT in the shared arsenal.

PLAYER TABLET

Teams see updated score/budget.
Learning remains observer-only.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Persist R2. Prepare R3 with 4 unspent attacks.	Do not restore R1/R2 attack cards.

11. ROUND 3 - RED ATTACK

AI assistant manipulation. Red chooses from what remains.

HOST SCREEN

FINAL ROUND
RED TEAM - ATTACK
Manipulate an AI assistant into taking an unauthorised action.
20 sec | Red budget $XXk

PLAYER TABLET

RED TABLETS
4 unspent Attack cards remain (assuming one used per prior round).
2 spent cards remain visible/disabled with team attribution.
Unaffordable remaining cards also disabled.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Use same attack component. No special Round 3 deck.	Winning attack charged once and marked SPENT. Three attack methods remain unused at game end.

12. ROUND 3 - BLUE DEFENDS

Same reusable Defence deck; scenario changes the best response.

HOST SCREEN

AI ASSISTANT UNDER ATTACK
Malicious instructions hidden in content try to trigger an unauthorised action.
BLUE: CHOOSE YOUR RESPONSE.

PLAYER TABLET

BLUE TABLETS
Same 6 Defence cards.
Best immediate response here: **Stop the Action**.
Cards remain reusable; budget still applies.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Run standard defence component.	Resolve final attack vs defence; apply final deltas.

13. FINAL RESOLUTION

End of three attack/defence rounds.

HOST SCREEN

FINAL ROUND RESULT
Selected attack vs selected defence.
SABOTAGE SUCCESSFUL / ATTACK BLOCKED
Final team scores + budgets calculated.

PLAYER TABLET

Teams see final round outcome.
Learning Mode remains observer-only until Task 4 ends.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Freeze Task 4 scores after result animation.	shared_attack_spent should contain exactly 3 attack IDs after normal completion.

14. TEAM RESULT + PAYOUT

Team play converts back to individual competition.

HOST SCREEN

AI CYBER SABOTAGE - FINAL
Red score / budget
Blue score / budget
Winning team revealed.
Then individual payout is calculated.

PLAYER TABLET

Competitor: team result + individual Task 4 share.
Learning Mode: no Task 4 points.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Reveal totals then payout.	Distribute each team's eligible Task 4 payout across its competing members; add to global individual score.

15. LEADERBOARD -> ULTIMATE SHOWDOWN

Teams dissolve.

HOST SCREEN

YOUR ALLIANCE IS OVER.
Red / Blue no longer matters.
Live leaderboard updates.
THE ULTIMATE SHOWDOWN
3 final questions. One winner.

PLAYER TABLET

Competitors return to individual mode + see new rank.
Learning Mode exits Task 4 observer lock.

DEVELOPER BEHAVIOUR	STATE / SCORING LOGIC
Clear team state; retain global score/rank.	Launch Task 5 with exactly 3 questions.

16. DEVELOPER STATE MAP + DATA FLAGS

The single-use Attack rule must live in shared game state, not team-local state.

T4_INIT	eligible competitors; Red/Blue split; budgets=100k; shared_attack_spent=[]
TEAM_REVEAL	10-second physical grouping
R1_ATTACK	Red votes among unspent + affordable attacks
R1_DEFEND	Blue votes among reusable affordable defences
R1_RESOLVE	charge once; mark attack spent globally; score
R2_ATTACK	Blue sees 1 spent + 5 unspent attack methods
R2_DEFEND	Red gets all reusable defences subject to affordability
R2_RESOLVE	mark second attack spent globally
R3_ATTACK	Red sees 2 spent + 4 unspent attack methods
R3_DEFEND	Blue gets reusable defence deck
R3_RESOLVE	mark third attack spent globally; freeze team result
TEAM_RESULT	show team scores / budgets / winner
PAYOUT	convert team performance to individual score
LEADERBOARD	re-sort individual competition
T5_TRANSITION	clear team state; launch 3-question Ultimate Showdown

Card status priority: SPENT overrides affordability for Attack cards. An Attack card already used stays SPENT regardless of budget. For unspent cards, disable when cost exceeds the acting team's budget.

Configuration: scenario copy, card costs, rewards, penalties, effectiveness matrix, timers and learning messages should remain configurable rather than hard-coded.